INDIA.RUNS

Build what next India runs on

Track 1 AI Systems & Workflow Innovation Challenge

Redrob Context & Ideathon Scope

Redrob AI is building an AI-native ecosystem spanning hiring, productivity, communication, discovery, workflows, recommendations, automation, and intelligent user experiences.

Participants are encouraged to think beyond standalone applications and explore how their ideas can:

- Extend existing Redrob capabilities
- Improve experiences for Redrob users
- Introduce new AI-native workflows
- Strengthen ecosystem engagement and network effects

Important

Participants are not expected to build a completely new platform from scratch.

Strong submissions will demonstrate:

- How the idea leverages existing Redrob capabilities
- What new capability, workflow, or value layer is introduced
- How the idea strengthens the overall Redrob ecosystem
- Why Redrob is uniquely positioned to enable this experience

Team Name : Tech Adi

Problem Statement : Recruiters spend 60%+ of time manually screening candidates.
Keyword-based ATS misses qualified candidates; Redrob signals are underutilised.
There is no autonomous intelligence layer to rank profiles at scale.

Team Members : Aditya Nadamuni (Solo Participant)

Problem Definition

Questions

- PROBLEM: Recruiters manually review 200+ profiles per role — slow, biased, and inconsistent.

75% of qualified candidates are eliminated by keyword-based ATS before a human reads them.
Keyword stuffing is rampant: candidates game the system, polluting the talent pool.

WHO EXPERIENCES THIS: Recruiters and HR teams on Redrob, and the candidates who deserve better.

WHY CURRENT APPROACH IS INSUFFICIENT:
- Keyword matching ignores verified behavioral signals (response rate, GitHub, assessments)
- No explainability: recruiter doesn't know why a profile was ranked higher
- Manual screening is not scalable past ~50 profiles per recruiter per day

WHY NOW: Multi-agent AI frameworks have reached production-grade reliability.
Redrob already stores unique behavioral signals no external ATS has access to.

Opportunity & Vision

Questions

- MARKET: $200B+ global recruiting market shifting to AI-native tooling.

Time-to-hire is the top KPI — AI can compress it from weeks to hours.

WHY THIS IS IMPORTANT:
- Redrob has 100K+ profiles enriched with redrob_signals unavailable anywhere else
- Behavioral data (response rates, assessments, GitHub) is a defensible moat
- Competitors can only see resume text; Redrob sees actual engagement patterns

FUTURE STATE WE ARE ENABLING:
RecruitMind — an AI recruiting co-pilot that does the work of a senior recruiter:
- Understands JD intent from free text in seconds
- Ranks 100K candidates using verified signals, not just keywords
- Explains every ranking decision with evidence from career history
- Drafts personalised outreach emails per candidate automatically
- Tracks replies and schedules interviews autonomously

Solution Overview

Questions

- SOLUTION: RecruitMind — Autonomous Multi-Agent Recruiting Intelligence for Redrob

AI-NATIVE (not AI-assisted): Agents make decisions end-to-end; recruiter only approves.

5 SPECIALISED AGENTS:
1. JD Parser — extracts required skills, seniority tier, domain from free-text JD
2. Candidate Ranker — scores 100K profiles in <30s using hybrid rule + signal engine
3. Fit Explainer — generates human-readable match reasoning per candidate
4. Outreach Composer — writes personalised email from candidate career history
5. Pipeline Manager — tracks replies, schedules interviews, updates pipeline status

REDROB CAPABILITIES LEVERAGED: Candidate DB, redrob_signals, job listings, messaging, calendar.

PROOF OF CONCEPT ALREADY BUILT: Core ranking engine validated on full 100K dataset.
Runtime: ~30s CPU | RAM: ~200MB | No GPU | 100% offline inference
Live demo: https://theadityanvs-redrob-ranker.streamlit.app

User Journey / Workflow Diagram

Questions

- INTERACTION: Recruiter pastes a Job Description — system triggers the pipeline automatically.

Within 60 seconds: ranked shortlist with per-candidate fit explanations appears on dashboard.
Recruiter approves candidates; personalised outreach sent automatically.
Pipeline Manager tracks replies and surfaces interview scheduling prompts.

INFORMATION FLOW: JD text -> JD Parser -> Candidate Ranker (100K profiles) -> Top 100 Shortlist -> Fit Explainer -> Outreach Composer -> Pipeline Manager -> Calendar.

AI TOUCHPOINTS: NLP parsing at input; scoring at ranking; LLM generation at outreach.

Mandatory Visual

- [Recruiter posts Job Description]

```
        |
        v
[ JD Parser Agent ]
extracts: skills[], seniority, domain
        |
        v
[ Candidate Ranker ]
```

AI Logic & Decision Flow

Questions

- AI INTERVENTION POINTS:

1. JD Parsing — NLP extracts structured requirements from free text
2. Candidate Scoring — weighted arithmetic over 6 verified components (no hallucination)
3. Fit Explanation — template-driven NL generation from scored components
4. Outreach — LLM (Phi-3 mini, CPU) personalises email from career history

HOW DECISIONS ARE MADE: All scoring is rule-based and auditable.
Weights: title+career 35% | skill trust 20% | behavioral 20%
experience 10% | availability 10% | education 5%
Honeypot multiplier: 1.0 (clean) or 0.05 (fraud pattern detected)

AGENT INTERACTION: Each agent reads/writes structured JSON to Redrob data layer.
No agent has side effects on another's computation — fully parallelisable.

Mandatory Visual

- [JD Free Text]

```
      |
      v
[ JD Parser: skills[], seniority_tier, domain ]
      |
      v
[ Candidate Stream (JSONL, 100K profiles) ]
```

System Architecture

Questions

- COMPONENTS: 5 agent microservices + Redrob Data Layer + FastAPI orchestrator + Celery queue.

SERVICE INTERACTIONS: Event-driven via Redrob internal event bus.
Each agent publishes/subscribes — no tight coupling.

REDROB SYSTEMS USED: Candidate DB, redrob_signals store, job listings API,
messaging platform (outreach delivery), calendar API (interview scheduling).

DEPLOYMENT: Containerised Docker; horizontally scalable Celery worker pool.
No new database — reads/writes to existing Redrob candidate store.

Mandatory Visual

-
```
      [ RECRUITER DASHBOARD - Redrob Web UI ]
          | REST API
   [ AGENT ORCHESTRATOR - FastAPI ]
     |     |      |     |
 [JD Parser][Ranker][Explainer][Outreach]
     |     |      |     |
 _____|________|__________|_________|_____
   |      REDROB DATA LAYER        |
```

Data, Context & Intelligence Layer

Questions

- DATA SOURCES: Redrob's existing candidate store — no external data needed.

CONTEXT RETRIEVAL: Streaming JSONL; ~200MB peak RAM for 100K profiles (no full-load).

HOW REDROB DATA ENABLES INTELLIGENCE:
  - career_history text   -> semantic career signal scanning
  - skills proficiency   -> trust-weighted skill scoring
  - redrob_signals   -> behavioral intelligence (unique to Redrob)
  - education tier   -> academic quality signal
  - open_to_work, notice -> real-time availability intelligence

STORAGE: Ranked output persisted as structured JSON per JD in Redrob DB.
No new database or storage system required.

Mandatory Visual

- DATA INPUTS

```
|
|-- candidate profile
|    |-- current_title, career_history  --> title_career scoring
|    |-- years_of_experience             --> experience curve
|    `-- location, willing_to_relocate   --> availability
|
```

Scalability & Technical Feasibility

Questions

- IMPLEMENTATION: Python 3.11 + FastAPI + Celery workers. Runs on existing Redrob cloud infra.

SCALABILITY:
- Current: 100K candidates in ~30s on a single CPU thread
- With 8 Celery workers: <5s (linear parallelism; stateless scoring)
- Trivially sharded by candidate_id range for larger datasets
- No GPU, no external API calls during ranking

TECHNICAL CHALLENGES & SOLUTIONS:
1. JD quality varies -> structured NLP extraction + keyword fallback
2. Keyword stuffing -> title+career text as primary signal (not skills list)
3. Profile fraud -> 3-rule honeypot detector (0.05x score penalty)
4. Outreach on CPU-only -> Phi-3 mini (3.8B, INT4 quantised) for personalisation

FEASIBILITY: Core ranking engine is production-validated on full 100K dataset today.
Outreach + Pipeline agents are modular additions; independently deployable.

Redrob Ecosystem Integration

Questions

- EXISTING CAPABILITIES LEVERAGED:

- Candidate database (100K+ profiles enriched with redrob_signals)
- Job listings (input source for JD Parser agent)
- Messaging platform (delivery channel for Outreach Composer)
- Calendar API (interview scheduling by Pipeline Manager)
- redrob_signals (recruiter_response_rate, interview_completion, github_activity)

NEW CAPABILITY ADDED: Autonomous ranking + explainability + outreach — all in one loop.
RecruitMind is a native Redrob plugin; no external SaaS subscription required.
Redrob becomes not just a candidate database but an active hiring intelligence platform.

Mandatory Visual

- REDROB PLATFORM

```
 ________________________________
|                    |
|  EXISTING        NEW        |
|                    |
|  Candidate DB  --> Ranker      |
```

Impact & Success Metrics

Questions

- QUANTITATIVE IMPACT:

- Time-to-shortlist: 3-5 days (manual) -> <10 minutes (RecruitMind)
- Profiles reviewed/hr: 5x increase per recruiter
- Candidate quality: NDCG@10 > 0.85 on internal validation dataset
- Fraud elimination: 100% of honeypot profiles detected and penalised

BUSINESS VALUE:

- $50K+ annual time savings per recruiter
- Higher offer-acceptance rate via personalised outreach (+23% industry avg)
- Reduced recruiter burnout; focus shifts to relationship-building

SUCCESS KPIs TRACKED IN REDROB DASHBOARD:

- Shortlist-to-interview conversion rate
- Offer-acceptance rate (per JD)
- Time-to-hire (JD posting to signed offer, in days)
- Recruiter NPS score (measured monthly)

Future Roadmap

Questions

- PHASE 1 - Q3 2026: Production rollout. A/B test vs. manual screening.

Agents deployed: JD Parser + Candidate Ranker + Fit Explainer (already built).

PHASE 2 - Q4 2026: Outreach Composer + Pipeline Manager agents launched.
Scoring weights fine-tuned on Redrob historical hire/reject outcome data.

PHASE 3 - Q1 2027: Multi-modal signals.
GitHub repo quality analysis, portfolio parsing, video interview sentiment (opt-in).

PHASE 4 - Q2 2027: Fully autonomous pipeline mode.
JD-to-shortlist-to-offer with fewer than 2 recruiter touchpoints.

3-YEAR VISION:
Redrob becomes the definitive AI-native hiring platform.
From discovery through onboarding — fully intelligent, fully explainable.
The redrob_signals dataset becomes the defensible proprietary AI moat.

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THANK YOU